**Higher National Diploma in Quantity Surveying**
**2nd Year, Second Semester Examination - 2019**
**QS 22073 - Structural Design and Detailing**

**Instructions for Candidate:**
Allow to use BS 8110 and BS 5268 codes
Answer five (05) Questions only

**No. of Questions : 06**
**No. of Pages : 04**
Time : 03 (hours)

1. A simply supported reinforced concrete rectangular beam is used to carry a dead load of 10kN/m (including self-weight of the beam) and live load of 12.5kN/m.
   Assume that,
   - No redistribution of loads, $\beta_b = 1$
   - Characteristic material strengths are
     $f_{cu}$ (concrete)= 25 N/mm$^2$ ,
     $f_y$ (main steel) = 460 N/mm$^2$
   - The cover to the stirrups = 25mm
   - Support width = 400mm
   - Clear span between supports= 7.0m
   - Overall depth of beam = 500mm

   Design the bending reinforcements of the above beam and do the relevant checks. Mention clearly the references of the code and assumptions made by you. (Total 20 Marks)

2. A Roof slab of a building has to be designed with four equal panels supported on beams as in fig 1 with 3.5m effective spans. The size of each panel is 3.5m X 9.0m. The cover to the tension reinforcement of the slab is 20mm. Design reinforcements at a middle support and a middle span of the roof slab by considering the finishes of the slab is 1kN/m$^2$ and Imposed load on the slab is 3.5kN/m$^2$. Check for deflection, cracks and shear and mention clearly all the references of code and assumption made by you.
   Assume that,
   - Density of concrete = 24kN/m$^3$.
   - Characteristics material strengths are, fcu (Concrete)= 30N/mm$^2$ , fy (Steel) = 460N/mm$^2$

   (Total 20 Marks)

3. A flanged beam is subjected to a sagging moment of 200kNm. Width of the flange is 600mm and width of the web is 300mm. Thickness of the flange is 150mm and overall height of the section is 550mm. Design main reinforcements and mention clearly the references in the code and assumptions made by you.

   Assume that,
   i. Cover to the main reinforcements = 25mm
   ii. Characteristics strengths of materials are $f_{cu}$ =30 N/mm$^2$ and $f_y$ = 460 N/mm$^2$

   (Total 20 Marks)

4. The reinforced concrete columns of size of 300 x 300 mm are used to support a 125mm thick one way spanning roof slab. The column is connected to the footing at the bottom and to the 250 x 400 mm beam at the top. The column height from footing to soffit of the beam is 3.00m. The design load of 1100kN is applied to an interior column axially without any eccentricity. If the frame is braced ,

   i. Classify the above interior column as short or slender (10 marks)

   ii. Determine the main reinforcement required for the above column to resist the axial load. $f_{cu} = 25\ N/mm^2$ , $f_y = 460\ N/mm^2$ (10 marks)

(Total 20 Marks)

5.

   i. Briefly discuss the advantages and disadvantages of steel as a structural material. (06 marks)

   ii. Design a lap joint between plates 65mmX8mm plate as shown in Fig(2), so as to transmit a factored load of 45kN using black bolts of 16mm diameter and grade 4.6. The plates are made of S275. (14 marks)

(Total 20 marks)

6. A timber beam of 3.0m length spans over an opening (Clear distance) and supports on 220mm thick brick wall at the ends which exerts long-duration loading of 3kN/m uniformly distributed design load. Check whether 100x250mm teak (D70) timber section is satisfied in the wet exposure condition. (Total 20 Marks)

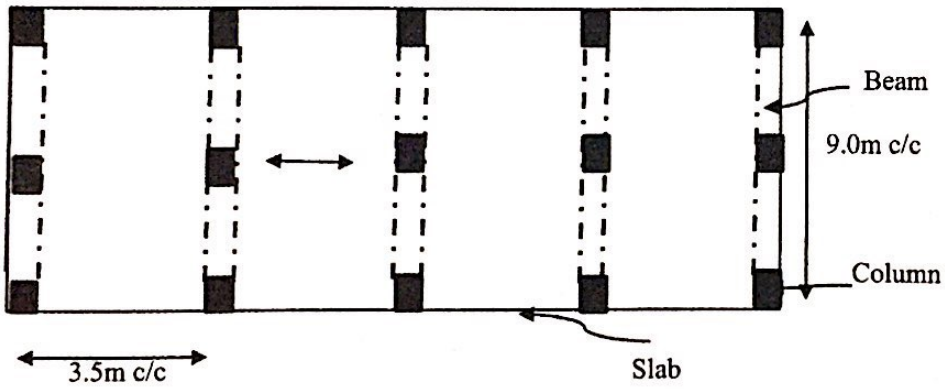


Fig 1

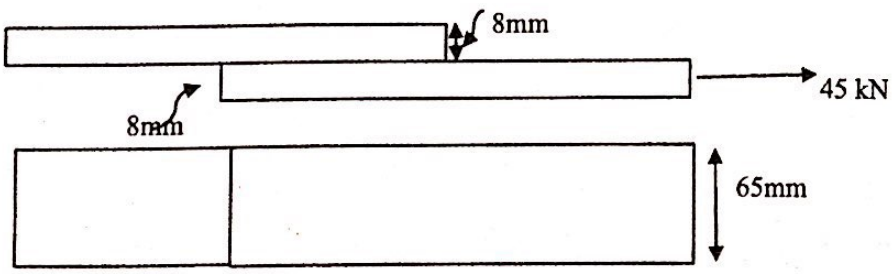


Fig 2

**Shear strength of bolts**

| Bolt Grade | Shear strength , $p_s$ (N/mm$^2$) |
|---|---|
| 4.6 | 160 |
| 8.8 | 375 |
| 10.9 | 400 |

**Bearing Strength of Bolts**

| **Bolt Grade** | **Bearing Strength** |
|---|---|
| **4.6** | 460 |
| **8.8** | 1000 |
| **10.9** | 1300 |

**Bearing Strength of Connected plates**

| **Steel Grade** | **S275** | **S355** | **S460** |
|---|---|---|---|
| **Bearing Strength(N/mm$^2$)** | 460 | 550 | 670 |

Bearing Capacity of bolt, $P_{bb} = dt_b p_{bb}$

Bearing capacity of connected plates, $P_{bs} = k_{bs} d t_p p_{bs}$ but $P_{bs} \leq 0.5 k_{bs} e t_p p_{bs}$

$k_{bs} = 1$ for Standard clearance holes

e = End distance

d = Nominal diameter of bolt

$t_p$ = Thickness of the connected plates

**Tensile stress area**

| Nominal size and tread diameter (mm) | Tensile stress area (mm$^2$) |
|---|---|
| 15 | 84.5 |
| 16 | 157 |
| 20 | 245 |
| 22 | 303 |
| 24 | 353 |